



沐燕舟

UNIST InnoCORE 博士后研究员
研究方向: AI 基础设施质量保障
个人主页: muyanzhou96.github.io
ORCID: 0000-0003-1816-2246

UNIST 计算机科学与工程系
邮箱: 602022320006@smail.nju.edu.cn
Google Scholar: [Profile](#)
研究领域: 软件测试、SE for AI、AI for SE

教育与工作经历

- **UNIST**
计算机科学与工程系
合作导师: *Mijung Kim*
InnoCORE 项目博士后研究员 2026.06-2026.11
- **南京大学**
软件学院, 南京
导师: 陈振宇、房春荣 专业: 软件工程
软件工程博士 2022.09-2026.06
- **天津大学**
智能与计算学部
导师: 王赞、刘爽、陈俊洁 专业: 软件工程
硕士 2019.09-2022.01
- **南通大学**
计算机科学与技术学院
导师: 陈翔 专业: 计算机科学与技术
学士 2015.09-2019.06

论文发表

2.1 第一作者论文

- **Yanzhou Mu**, Shuo Meng, Mijung Kim, Xiang Chen, Chunrong Fang, Zhenyu Chen, Juan Zhai. CIPIHunter: Detecting Configuration-Induced Prediction Instability in Deep Learning Frameworks. In Proceedings of IEEE/ACM Automated Software Engineering Conference (ASE 2026). ACM, Munich, Germany. CCF-A.
- **Yanzhou Mu**, Rong Wang, Juan Zhai, Chunrong Fang, Xiang Chen, Jiacong Wu, An Guo, Jiawei Shen, Bingzhuo Li, and Zhenyu Chen. Understanding LLM-Centric Challenges for Deep Learning Frameworks: An Empirical Analysis. ACM Transactions on Software Engineering and Methodology, 2026. CCF-A.
- **Yanzhou Mu**, Rong Wang, Juan Zhai, Chunrong Fang, Xiang Chen, Peiran Yang, Zhixiang Cao, Ruixiang Qian, Shaoyu Yang, and Zhenyu Chen. Deep Learning Framework Testing via Model Mutation: How far are we? IEEE Transactions on Software Engineering, 2025. CCF-A.
- **Yanzhou Mu**, Juan Zhai, Chunrong Fang, Xiang Chen, Zhixiang Cao, Peiran Yang, Kexin Zhao, An Guo and Zhenyu Chen. Improving Deep Learning Framework Testing with Model-Level Metamorphic Testing. International Symposium on Software Testing and Analysis (ISSTA 2025), Thondheim, Norway, 2025. CCF-A.
- **Yanzhou Mu**, Juan Zhai, Chunrong Fang, Xiang Chen, Zhixiang Cao, Peiran Yang, Yinglong Zou, Tao Zheng, and Zhenyu Chen. DevMuT: Testing Deep Learning Framework via Developer Expertise-Based Mutation. In Proceedings of IEEE/ACM Automated Software Engineering Conference (ASE 2024). ACM, Sacramento, California, United States. CCF-A.
- **沐燕舟**, 王赞, 陈翔, 陈俊洁, 赵静珂, 王建敏. 采用多目标优化的深度学习测试优化方法 / Deep learning test optimization method using multi-objective optimization. 中文版本: 软件学报, 2022, 33(07):2499-2524, DOI:10.13328/j.cnki.jos.006 英文版本: International Journal of Software & Informatics, 12(4), 2022.
- **Yanzhou Mu**, Zan Wang, Shuang Liu, Jun Sun, Junjie Chen, and Xiang Chen. HARS: Heuristic-enhanced adaptive randomized scheduling for concurrency testing. In 2021 IEEE 21st International Conference on Software Quality, Reliability and Security (QRS), pages 219-230. IEEE, 2021.

2.2 合作论文

- Yuan Xiao, Yuchen Chen, Jiaming Wang, Wei Song, Jun Sun, Shiqing Ma, **Yanzhou Mu**, Juan Zhai, Chunrong

Fang, Jin Song Dong, and Zhenyu Chen. Train in Vain: Functionality-Preserving Poisoning to Prevent Unauthorized Use of Code Datasets. Findings of the Association for Computational Linguistics, 2026. CCF-A.

- Xinxue Zhu, Jiacong Wu, Xiaoyu Zhang, Tianlin Li, **Yanzhou Mu**, Juan Zhai, Chao Shen, Chunrong Fang, and Yang Liu. An Empirical Study of Bugs in Modern LLM Agent Frameworks. The 2nd International Workshop on Large Language Model Supply Chain Analysis (LLMSC 2026), co-located with FSE 2026, 2026. 已接收.
- An Guo, Shuoxiao Zhang, Enyi Tang, Xinyu Gao, Haomin Pang, Haoxiang Tian, **Yanzhou Mu**, Wu Wen, Chunrong Fang, and Zhenyu Chen. When Autonomous Vehicle Meets V2X Cooperative Perception: How Far Are We? The 40th IEEE/ACM International Conference on Automated Software Engineering, November 16-20, 2025, Seoul, South Korea. CCF-A.
- Chunyu Zhao, **Yanzhou Mu**, Xiang Chen, Jingke Zhao, Xiaolin Ju, and Gan Wang. Can test input selection methods for deep neural network guarantee test diversity? A large-scale empirical study. Information and Software Technology, 150:106982, 2022. 共同一作.
- Xiang Chen, **Yanzhou Mu**, Ke Liu, Zhanqi Cui, and Chao Ni. Revisiting heterogeneous defect prediction methods: How far are we? Information and Software Technology, 130:106441, 2021. 共同一作.
- Xiang Chen, **Yanzhou Mu**, Yubin Qu, Chao Ni, Meng Liu, Tong He, and Shangqing Liu. Do different cross-project defect prediction methods identify the same defective modules? Journal of Software: Evolution and Process, 32(5):e2234, 2020. 共同一作.

2.3 早期论文

- Hao Li, Xiaohong Li, Xiang Chen, Xiaofei Xie, **Yanzhou Mu**, and Zhiyong Feng. Cross-project defect prediction via asttoken2vec and blstm-based neural network. In 2019 International Joint Conference on Neural Networks (IJCNN), pages 1-8. IEEE, 2019.

2.4 在审稿件与预印本

- Zhixiang Cao, Di Tian, Runwei Guan, **Yanzhou Mu**, Xiaolou Sun, Shaofeng Liang, Daizong Liu, Tao Huang, Yutao Yue, Henghui Ding, Bin Fang, Alex Zhou, Qing-Long Han, and Hui Xiong. Tactile-based Multimodal Fusion in Embodied Intelligence: A Survey of Vision, Language, and Contact-Driven Paradigms. arXiv preprint, under review, 2026. DOI:10.48550/arXiv.2605.17336. 预印本.
- Yilong Zou, Juan Zhai, Chunrong Fang, **Yanzhou Mu**, Jiawei Liu and Zhenyu Chen. Deep Learning Framework Testing via Heuristic Guidance Based on Multiple Model Measurements, ACM Transactions on Software Engineering and Methodology, 2025, CCF-A, **Major revision**.
- An Guo, Xinyu Gao, Chunrong Fang, Haoxiang Tian, Weisong Sun, **Yanzhou Mu**, Shuncheng Tang, Lei Ma, Xiapu Luo, and Zhenyu Chen. Generate Realistic Test Scenes for V2X Communication Systems. ACM Transactions on Software Engineering and Methodology, 2025, CCF-A, **Major revision**.

项目经历

- **MindSpore Model Generalization Testing Technology | (已完成, 学生负责人)**
 - 开发了基于国产深度学习框架 MindSpore 的模型结构泛化测试工具。该工具提取主流 CV 与 NLP 模型结构的特征因子、取值范围和约束, 修改模型结构并执行测试以发现缺陷。项目成果作为重要支撑, 帮助合作团队获得华为 2023 年度创新测试应用奖。
- **Research on Key Mutation Testing Techniques for DL Frameworks | (进行中, 学生负责人)**
 - 设计模拟真实用户开发操作的变异规则, 生成贴近实际场景的测试模型; 结合框架静态与动态特征评估测试数据的多样性和充分性; 采用启发式方法指导变异测试, 并结合经验回放提升测试效率与质量。项目成果发表于 ASE 2024, 相关发明专利申请正在进行中。

- **Fast Detection, Localization, and Patch Generation for Concurrency Defects | (已完成, 项目参与人)**
 - 设计用于优化 Java 并发程序测试的启发式调度测试方法, 提升缺陷检测效率与效果。项目成果发表于 QRS 2021, 并获授权发明专利。
- **Generalization Testing for Large-Model Frameworks | (已完成, 学生负责人)**
 - 开发了面向大模型训练系统的泛化测试框架, 将模型分解为可复用的结构模块, 在模型结构、并行策略与优化策略的联合空间中生成语义有效的模型变体。进一步研究横跨模型结构、并行、优化与后端层的配置约束, 区分语义约束与实现层限制, 并在保持变异意图的前提下推进深层分布式执行。基于 PTA 与 MindSpore 系框架开展跨框架差分测试。累计产出 52 份问题报告, 其中 44 项被归类为高可信度疑似框架缺陷。

专利

- **Yanzhou Mu**; Zan Wang; Shuang Liu. A heuristic rule-based concurrent adaptive random testing method. Patent No: CN113468047A.
- Junjie Chen; **Yanzhou Mu**; Zan Wang; Jianmin Wang; Jiao Jia. A multi-objective optimization-based method for selecting deep learning test inputs. Patent No: CN114721934A.
- Zhenyu Chen; Yidong Ke; Jiawei Liu; **Yanzhou Mu**. A neural network fuzz testing method guided by mutation image entropy values. Patent No: CN117218051A.
- Yongping Liu; Zhenyu Chen; Chunrong Fang; **Yanzhou Mu**; Ruifa Luo; Shuiyong Du. Mutation testing method, device, and computer equipment based on neuron characteristics of intelligent traffic models. Patent No: CN118689772A.
- Yongping Liu; Zhenyu Chen; Chunrong Fang; **Yanzhou Mu**; Ruifa Luo; Shuiyong Du. Domain-specific mutation testing method, device, and computer equipment based on intelligent traffic V2X models. Patent No: CN118689771A.
- Zhijin Guan; Haiying Ma; Hehe Gu; Zongyuan Zhang; **Yanzhou Mu**; Jing Zhao; Meng Liu. A quantum circuit image recognition method. Patent No: CN106997462A.

奖励与荣誉

• 南京大学江苏银行奖学金	2024.10
• 华为 2023 年度创新测试应用奖	2023.11
• 天津大学优秀毕业生	2022.06
• 天津大学优秀共青团员	2021.01
• 南通大学 2019 届优秀毕业设计 (论文)	2019.06
• 江苏省优秀学生干部	2018.05
• 国家奖学金 (本科生)	2017.11
• 南通市三好学生	2017.05
• 南通大学三好学生标兵	2016.09